

Quantile Regression

ECMT 676 (Econometric II) Lecture Notes

TAMU

April 7, 2014

Why Quantile Regression?

- Often want to know the effect of changing a covariate – such as a policy intervention – on features of the distribution other than the mean.
- For example, how does eligibility in a particular kind of pension plan affect total wealth at different quantiles of the wealth distribution? The mean effect, while useful, may mask very different effects in different parts of the wealth distribution.

- Another reason to focus on the median (and quantile more generally) is that sometimes we can estimate parameters in underlying models under weaker assumptions using zero conditional median restrictions, rather than zero conditional mean restrictions.
- An important case is data censoring, which we cover later.
- Manipulations with medians (and quantiles) are useful for certain corner solution models, too.

- When we apply different estimation methods (say, ordinary least squares and least absolute deviations) to the same linear model, we must remember that these methods generally identify *different quantities* (mean versus median in this case).
- In the statistics literature the focus on LAD has often been on its resistance to outliers (which it certainly has). But there are other reasons OLS and LAD can give different results; it need have nothing to do with extreme data points.

Review of Means, Medians, and Quantiles

- Start with a linear population model, where β is $K \times 1$:

$$y = \alpha + \mathbf{x}\beta + u. \quad ((1))$$

- Assume $E(u^2) < \infty$, so that the distribution of u is not too spread out. (So, for example, we rule out a Cauchy distribution for u , or a t_2 distribution.)

- We call this equation a “linear model” because it is linear in its parameters.

There are many different ways to estimate the parameters of this model; the goal is to evaluate the quality of the estimation procedures under different assumptions.

- Ordinary Least Squares (OLS) Estimation:

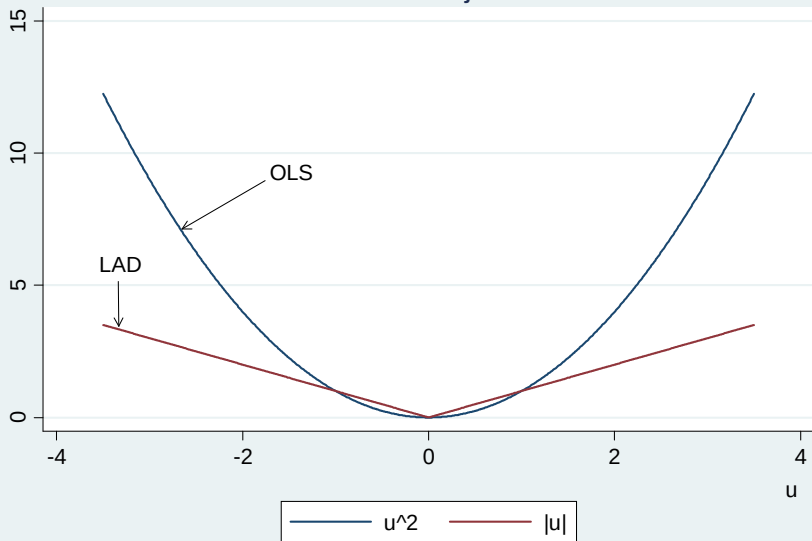
$$\min_{a, \mathbf{b}} \sum_{i=1}^N (y_i - a - \mathbf{x}_i \mathbf{b})^2. \quad ((2))$$

- Least Absolute Deviations (LAD) Estimation:

$$\min_{a, \mathbf{b}} \sum_{i=1}^N |y_i - a - \mathbf{x}_i \mathbf{b}|. \quad ((3))$$

- We should **not** now refer to equation (1) as an “OLS model” or an “LAD model.” OLS and LAD are *estimation methods*, not “models.”

The OLS and LAD Objective Functions



- There are infinitely many objective functions we could use. For example, for given $r > 0$,

$$\min_{a, \mathbf{b}} \sum_{i=1}^N |y_i - a - \mathbf{x}_i \mathbf{b}|^r.$$

- Rarely see choices other than $r = 1$ or $r = 2$.
- Question: With a large random sample, when should we expect the **slope** estimates from OLS and LAD to be similar?
- There are two important cases.

(i) If

$D(u|\mathbf{x})$ is symmetric about zero ((4))

then OLS and LAD both consistently estimate α and β because, under (4),
 $E(u|\mathbf{x}) = \text{Med}(u|\mathbf{x}) = 0$.

- Consequently,

$$\begin{aligned} E(y|\mathbf{x}) &= \alpha + \mathbf{x}\beta \\ \text{Med}(y|\mathbf{x}) &= \alpha + \mathbf{x}\beta \end{aligned}$$

- As we know, OLS under random sampling consistently estimates the parameters in a *conditional mean*. Other than ruling out perfect collinearity in $\mathbf{x}$, sufficient moment conditions are $E(u^2) < \infty$ and $E(x_j^2) < \infty$
- Under fairly weak conditions, LAD consistently estimates the parameters in a *conditional median*.

(ii) If

$$u \text{ is independent of } \mathbf{x} \text{ with } E(u) = 0, \quad ((5))$$

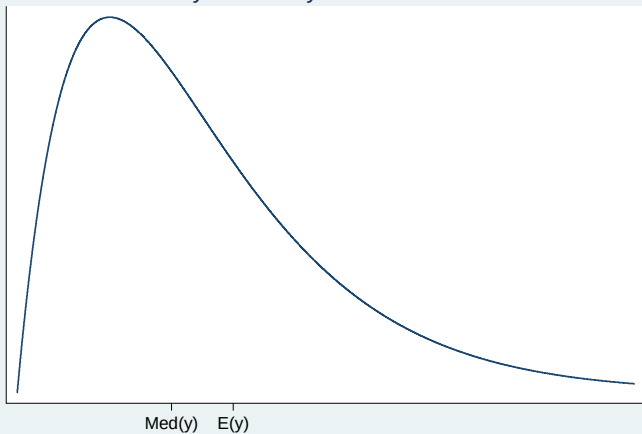
where $E(u) = 0$ is the normalization that identifies α , then OLS and LAD both consistently estimate the slopes, β .

- By (5), $E(u|\mathbf{x}) = 0$ and so we still have $E(y|\mathbf{x}) = \alpha + \mathbf{x}\beta$.
- If u has an asymmetric distribution, then $Med(u) \equiv \eta \neq 0$, and $\hat{\alpha}_{LAD}$ converges to $\alpha + \eta$ because

$$\begin{aligned} Med(y|\mathbf{x}) &= \alpha + \mathbf{x}\beta + Med(u|\mathbf{x}) \\ &= \alpha + \mathbf{x}\beta + \eta = (\alpha + \eta) + \mathbf{x}\beta. \end{aligned}$$

- In many applications, neither (4) nor (5) is likely to be true. For example, y may be a measure of wealth, in which case the error distribution is probably asymmetric and $Var(y|\mathbf{x})$ not constant.

Density for an Asymmetric Distribution



- Aside: Suppose $u \sim \text{Normal}(0, \sigma^2)$ and define $y = \exp(u)$, so that y has a lognormal distribution.
- Then $\text{Med}(y) = \exp[\text{Med}(u)] = \exp(0) = 1$ and $E[\exp(u)] = \exp(\sigma^2/2) > 1$. Therefore,

$$E(y) > \text{Med}(y),$$

as is often the case for asymmetric distributions that model economic variables.

- Important conclusion: If $D(u|\mathbf{x})$ is asymmetric and changes with $\mathbf{x}$, then we should not expect OLS and LAD to deliver similar estimates of β , even for “thin-tailed” distributions.
- Therefore, claims that substantive differences between OLS and LAD estimators must be due to “outliers” may be unwarranted.
- Of course, LAD is much more resilient to changes in extreme values because, as a measure of central tendency, the median is much less sensitive than the mean to changes in extreme values.
- OLS consistently estimates the parameters that provide the best mean-square approximation to $E(y|\mathbf{x})$. Unfortunately, characterizing LAD under misspecification is harder but possible (later).
- One case where LAD is clearly preferred is when $E|u| < \infty$ but $E(u^2) = \infty$.

- Advantage for median over mean: *median passes through monotonic functions.*
- For example, suppose $\log(y) = \alpha + \mathbf{x}\beta + u$ and $\text{Med}(u|\mathbf{x}) = 0$.
- Write $y = \exp(\alpha + \mathbf{x}\beta + u)$. Then

$$\begin{aligned}\text{Med}(y|\mathbf{x}) &= \text{Med}[\exp(\alpha + \mathbf{x}\beta + u)|\mathbf{x}] = \exp[\alpha + \mathbf{x}\beta + \text{Med}(u|\mathbf{x})] \\ &= \exp(\alpha + \mathbf{x}\beta).\end{aligned}$$

- The partial effect of x_j on $\text{Med}(y|\mathbf{x})$ has the same sign as β_j . If x_j is continuous,

$$\partial \text{Med}(y|\mathbf{x}) / \partial x_j = \beta_j \exp(\alpha + \mathbf{x}\beta).$$

- The pass-through does not hold for the mean generally.

- Generally

$$\text{Med}[g(y)|\mathbf{x}] = g[\text{Med}(y|\mathbf{x})],$$

where $g(\cdot)$ is monotonically increasing or decreasing (and need not be strictly increasing or decreasing) on the support of y .

- Later we will see that being able to pass the median through monotonic functions is very useful when data have been censored.
- We know that for nonlinear functions $g(\cdot)$,

$$E[g(y)|\mathbf{x}] \neq g[E(y|\mathbf{x})].$$

- The expectation cannot be passed through monotonic functions, but it has useful properties that the median does not: linearity and the law of iterated expectations.
- The median operator is not linear; in particular, $Med(x + y) \neq Med(x) + Med(y)$. There is no “law of iterated medians.”
- Consider $x, y \sim \text{bernoulli}(p)$, and $x \perp y$.

$$\begin{cases} P(x = 0) = 1 - p \\ P(x = 1) = p \end{cases}$$

- The distribution of $x + y$

$$\begin{cases} P(x + y = 0) = (1 - p)^2 \\ P(x + y = 1) = 2p - 2p^2 \\ P(x + y = 2) = p^2 \end{cases}$$

- So

$$\text{Med}(x + y) = \begin{cases} 0, & \text{if } p < 0.3 \\ 1, & \text{if } 0.3 < p < 0.7 \\ 2, & \text{if } p > 0.7 \end{cases}$$

using $\sqrt{0.5} \approx 0.7$.

- However,

$$\text{Med}(x) + \text{Med}(y) = 2\text{Med}(x) = \begin{cases} 0, & \text{if } p < 0.5 \\ 2, & \text{if } p > 0.5 \end{cases}$$

Quantiles

- For $0 < \tau < 1$, $q(\tau)$ is a τ^{th} quantile of y_i if $P[y_i \leq q(\tau)] \geq \tau$ and $P[y_i \geq q(\tau)] \geq 1 - \tau$.
- With this definition, a quantile need not be unique. (Special case: a median need not be unique.)
- Sometimes the quantile is defined uniquely as

$$q(\tau) = \{a : \min_{a \in \mathbb{R}} P(y_i \leq a) \geq \tau\}.$$

- Having a unique definition is not especially useful because the first definition is relevant for the subsequent optimization problem.

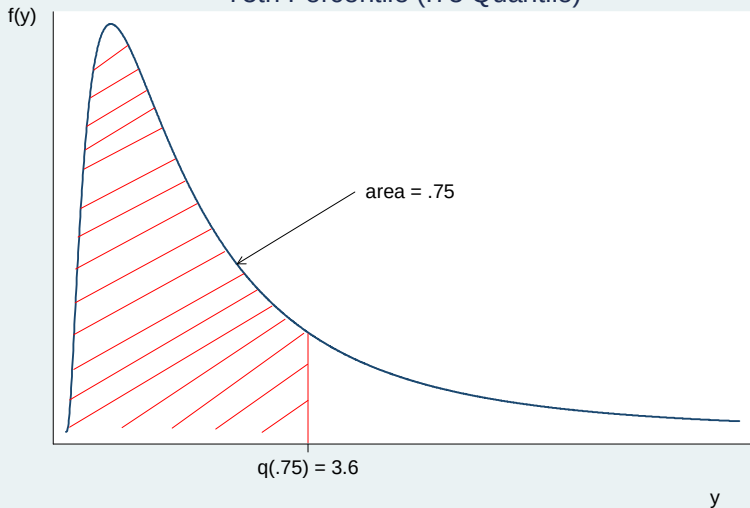
- When y_i is continuous with a strictly increasing CDF, $q(\tau)$ is the unique value such that

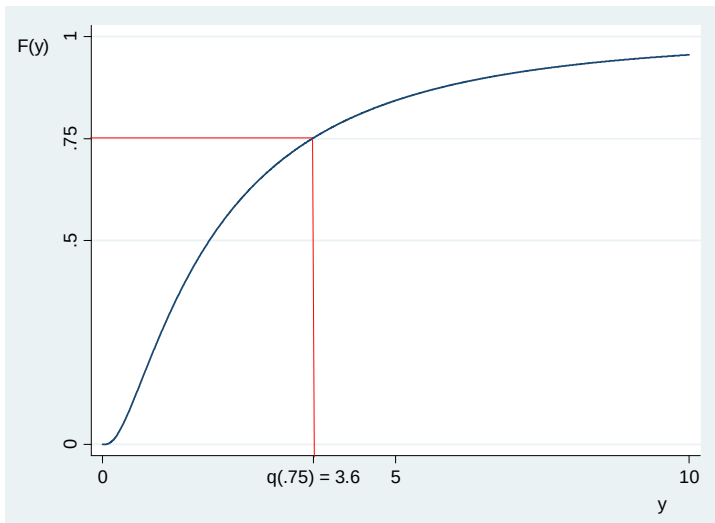
$$P[y_i \leq q(\tau)] = \tau$$

$$P[y_i \geq q(\tau)] = P[y_i > q(\tau)] = 1 - \tau.$$

- If $F_y(\cdot)$ is a strictly increasing CDF then $q(\tau) = F_y^{-1}(\tau)$.
- Non-uniqueness of quantiles arises when the CDF has flat spots. This is more likely when y_i is discrete.

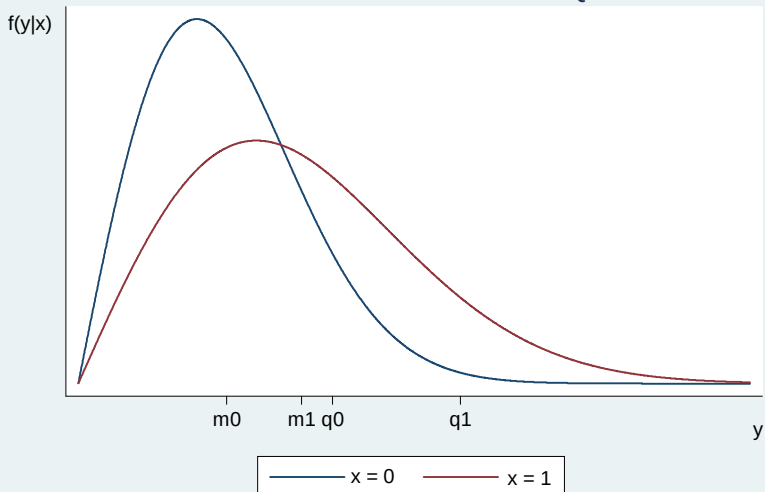
75th Percentile (.75 Quantile)





- In **quantile regression** we focus on conditional quantiles.
- Consider a case with a single binary explanatory variable, x . Then we can study two conditional distributions: $D(y|x = 0)$ and $D(y|x = 1)$.
- Let m_0 and m_1 be the means from the two distributions. Let q_0 and q_1 be the .90 quantiles from the two distributions. We want to know the change from q_0 to q_1 in addition to the change in m_0 to m_1 .
- With quantile regression, we can study the change in any quantile in the distribution.

Two Distributions: Means and .90 Quantiles



Quantile Regression

- More generally, we are interested in quantiles conditional on many covariates.
- Index the parameters by $\tau \in (0, 1)$.
- Assume that the conditional quantile function is unique and that it is linear:

$$Quant_{\tau}(y_i | \mathbf{x}_i) = \alpha(\tau) + \mathbf{x}_i \boldsymbol{\beta}(\tau). \quad ((9))$$

- To apply the analogy principle we need an objective function identifies the parameters $\alpha(\tau)$ and $\boldsymbol{\beta}(\tau)$.

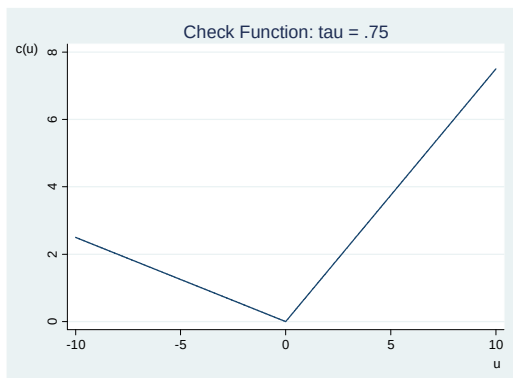
- Define the **check function** for $u \in \mathbb{R}$ as

$$\begin{aligned} c_{\tau}(u) &= (\tau 1[u \geq 0] + (1 - \tau) 1[u < 0]) |u| \\ &= (\tau - 1[u < 0]) u, \end{aligned}$$

where $1[\cdot]$ is the indicator function.

- Also called the **τ -absolute loss function**.
- When $\tau = 0.5$,

$$c_{0.5}(u) = |u| / 2$$



- The check function identifies $\alpha(\tau)$ and $\beta(\tau)$ in the sense that these parameters solve the population problem

$$\min_{\alpha \in \mathbb{R}, \beta \in \mathbb{R}^K} E[c_\tau(y_i - \alpha - \mathbf{x}_i\beta)].$$

- For identification we have to verify (or just assume) uniqueness.
- The proof proceeds by showing that, for any $\mathbf{x}_i$, $\alpha(\tau)$ and $\beta(\tau)$ solve the conditional problem,

$$\min_{\alpha \in \mathbb{R}, \beta \in \mathbb{R}^K} E[c_\tau(y_i - \alpha - \mathbf{x}_i\beta) | \mathbf{x}_i].$$

- Manski (1988, *Analog Estimation Methods in Econometrics*) contains a proof.

- The analogy principle leads to the sample analog:

$$\min_{\alpha \in \mathbb{R}, \beta \in \mathbb{R}^K} \sum_{i=1}^N c_{\tau}(y_i - \alpha - \mathbf{x}_i \beta), \quad ((10))$$

- Usually solved by linear programming.
- The solutions $\hat{\alpha}(\tau)$ and $\hat{\beta}(\tau)$ are the *quantile regression* (QR) estimators.

Quantile Crossing

- An attractive feature of quantile regression that has been emphasized is that it enables us to look at slices of the conditional distribution without any reliance on global distributional assumptions.
- Only local information near the specified quantile is employed.
- But we could find that estimated quantile function *cross* (thus violating the monotonicity property of quantiles), i.e.

for some x_i , $\hat{Q}_Y(\tau|x_i) = x_i'\hat{\beta}_\tau$ may not be monotone in τ

- Such crossing is typically in the outlying regions.

- The conditional quantiles at the sample mean must be monotone (Koenker, 2005, p.56)
- i.e. $\hat{Q}_Y(\tau|\bar{x}) = \bar{x}'\hat{\beta}_\tau$ must be monotone in $\tau \in [0, 1]$.
- If there is a significant number of observed points x at which $\hat{Q}_Y(\tau|x)$ is non-monotonic, then this can be taken as evidence of misspecification of the covariate effects.

Heteroskedasticity

- Consider a linear model

$$y_i = \beta_0 + \beta_1 x_i + (\gamma_0 + \gamma_1 x_i) v_i,$$

where v_i is iid with the CDF F , v_i is independent of x_i .

- If $\gamma_1 = 0$, then all conditional quantiles are linear and parallel.
- If $\gamma_1 \neq 0$, then conditional quantiles are generally nonlinear.
- Then

$$Q_{y_i}(\tau|x_i) = \begin{cases} \beta_0 + \beta_1 x_i + (\gamma_0 + \gamma_1 x_i) F^{-1}(\tau), & \text{if } \gamma_0 + \gamma_1 x_i \geq 0 \\ \beta_0 + \beta_1 x_i + (\gamma_0 + \gamma_1 x_i) F^{-1}(1 - \tau), & \text{if } \gamma_0 + \gamma_1 x_i < 0 \end{cases}$$

- So heteroskedasticity introduces nonlinearity to the conditional quantile function.

Heteroskedasticity

- A fact: Suppose v is a r.v. with the CDF F , or $Q_v(\tau) = F^{-1}(\tau)$, then

$$Q_{-v}(\tau) = -F^{-1}(1 - \tau).$$

- Proof. We only need to show $P(-v < -F^{-1}(1 - \tau)) = \tau$ or $P(v > F^{-1}(1 - \tau)) = \tau$, which is implied by $P(F(v) > 1 - \tau) = \tau$.
- The last equality follows from the fact that $F(v) \sim \text{Uniform}(0, 1)$.
- We only need $P(F(v) < u) = u$, which follows from

$$P(F(v) < u) = P(v < F^{-1}(u)) = P(v < Q(u)) = u.$$

An Application

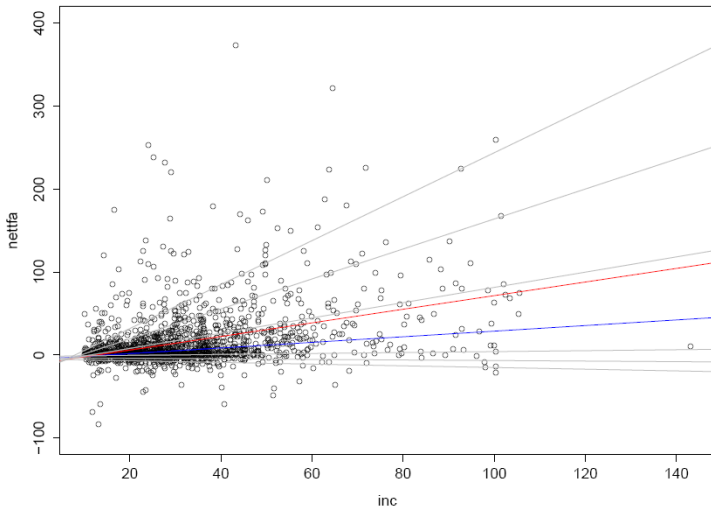
Individual-level data, all singles, $n = 2017$. (Abadie, 2003, JoE)

Source: Survey of Income and Program Participation (SIPP) of 1991

Dependent variables: Net total financial assets

Regressors: age, income, e401k

Net total financial assets vs. Income (a linear model)

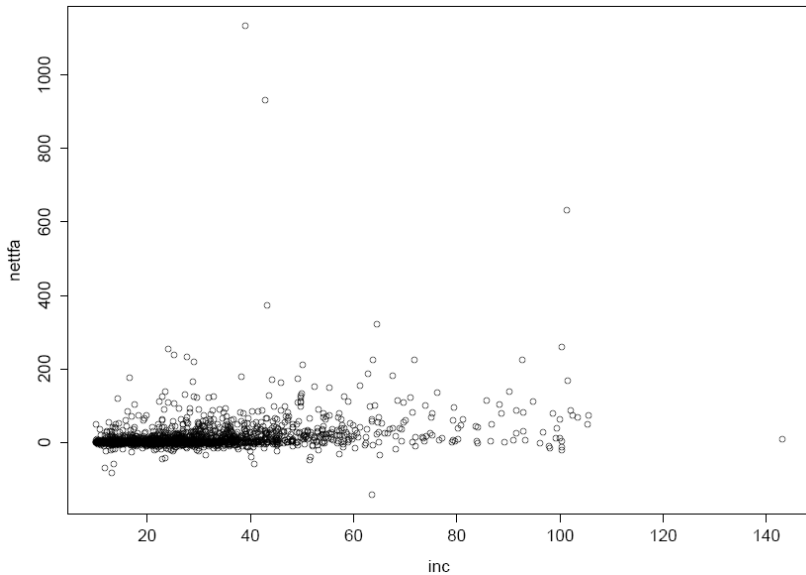


red: OLS

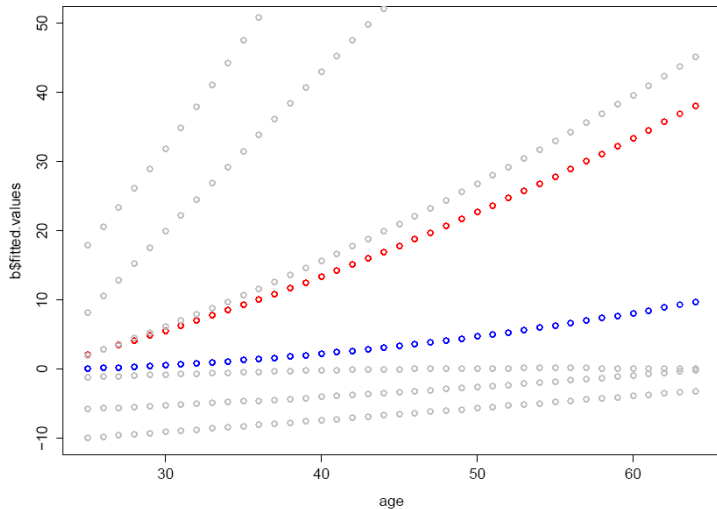
blue: Median regression

gray: 5%, 10%, 25%, 75%, 90%, 95%

Plot of all points (with outliers)



Net total financial assets vs. Age (a quadratic model)

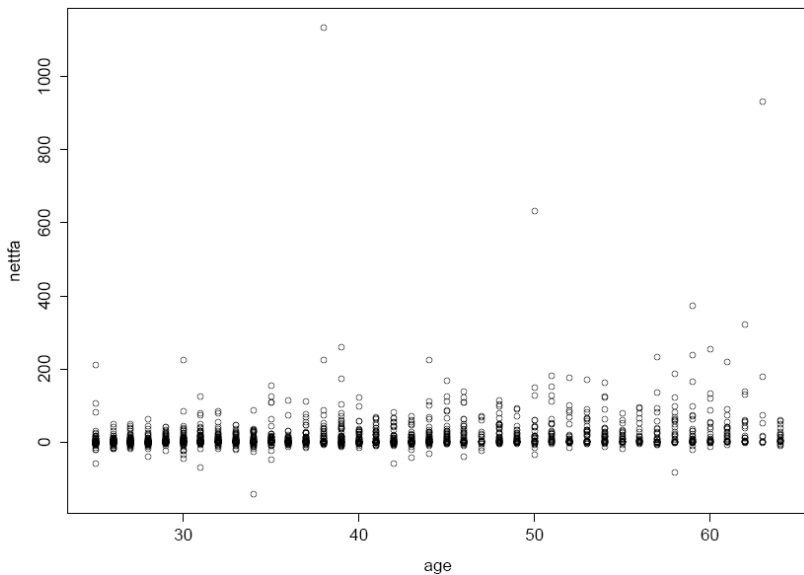


red: OLS

blue: Median regression

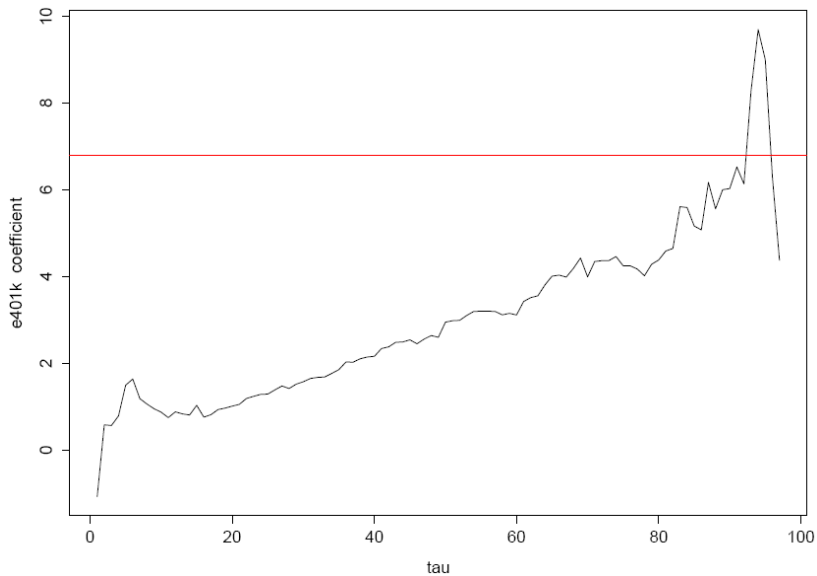
gray: 10%, 25%, 75%, 90%

Plot of all points (with outliers)



Now we consider a multivariate model.

$$\begin{aligned}\text{mean} &: \text{nettfa} = 2.5 + 0.8\text{inc} - 1.6\text{age} + 0.02\text{agesq} + 6.8\text{e401k} \\ \tau = 0.25 &: \text{nettfa} = -4.4 + 0.07\text{inc} + 0.03\text{age} + 0.00\text{agesq} + 1.3\text{e401k} \\ \tau = 0.50 &: \text{nettfa} = -3.6 + 0.3\text{inc} - 0.2\text{age} + 0.00\text{agesq} + 2.6\text{e401k} \\ \tau = 0.75 &: \text{nettfa} = 7.5 + 0.8\text{inc} - 1.4\text{age} + 0.02\text{agesq} + 4.5\text{e401k}\end{aligned}$$



Consistency

- Consistency of QR is fairly straightforward from general results on M-estimation.
- Because the objective function is *convex*, we can even drop the annoying assumption that the parameter space is compact (closed and bounded). See Newey and McFadden (Handbook).
- Often u_i is assumed to have a continuous distribution, conditional on $\mathbf{x}_i = \mathbf{x}$.
- The key is that purely discrete y_i is ruled out.
- As with OLS, need to rule out perfect collinearity in the population distribution of $\mathbf{x}$.

- The identification condition can be stated as: $E[f_u(0|\mathbf{x}_i)\mathbf{x}_i'\mathbf{x}_i]$ is nonsingular.
- If u_i is independent of $\mathbf{x}_i$ – a leading case, especially historically – then we need $f_u(0) > 0$ and $E(\mathbf{x}_i'\mathbf{x}_i)$ nonsingular.
- When u_i and $\mathbf{x}_i$ are not independent, we might impose a lower bound, $f_u(0|\mathbf{x}) \geq \rho > 0$, all $\mathbf{x}$, along with no perfect collinearity in the population distribution of $\mathbf{x}_i$.

Asymptotic Normality

- First assume the linear model is correctly specified for a given quantile τ :

$$y_i = \mathbf{x}_i \boldsymbol{\theta}_o + u_i, \quad Q_\tau(u_i | \mathbf{x}_i) = 0, \quad ((14))$$

where $x_1 \equiv 1$.

- By construction, $F_u(0 | \mathbf{x}) = \tau$ for all $\mathbf{x}$, where $F_u(\cdot | \mathbf{x})$ is the conditional CDF.
- For a random draw i , the objective function is

$$q(\mathbf{w}_i, \boldsymbol{\theta}) = (\tau - 1[y_i - \mathbf{x}_i \boldsymbol{\theta} < 0])(y_i - \mathbf{x}_i \boldsymbol{\theta})$$

- $q(\mathbf{w}_i, \cdot)$ is differentiable in most of $\boldsymbol{\theta}$, except $\boldsymbol{\theta}$ such that $y_i - \mathbf{x}_i \boldsymbol{\theta} = 0$ (at the kink).
- Such $\boldsymbol{\theta}$'s (non-differentiable points) are still infinitely many when $\boldsymbol{\theta}$ is multi-dimensional.

- Assume that $P(y_i - \mathbf{x}_i\boldsymbol{\theta} = 0) = 0$ for all $\boldsymbol{\theta}$.
- Note we can write $y_i - \mathbf{x}_i\boldsymbol{\theta} = u_i - \mathbf{x}_i(\boldsymbol{\theta} - \boldsymbol{\theta}_o)$. We can ensure $P(y_i - \mathbf{x}_i\boldsymbol{\theta} = 0) = 0$ by assuming u_i is a continuous random variable.
- Recall that $q(\mathbf{w}_i, \boldsymbol{\theta}) = (\tau - 1[y_i - \mathbf{x}_i\boldsymbol{\theta} < 0])(y_i - \mathbf{x}_i\boldsymbol{\theta})$.
- Define a score function as

$$\mathbf{s}(\mathbf{w}_i, \boldsymbol{\theta}) = -\mathbf{x}_i'(\tau - 1[y_i - \mathbf{x}_i\boldsymbol{\theta} < 0])$$

- Then

$$E[\mathbf{s}(\mathbf{w}_i, \boldsymbol{\theta}_o) | \mathbf{x}_i] = -\mathbf{x}_i'(\tau - \tau) = \mathbf{0},$$

where we used $E\{1[y_i - \mathbf{x}_i\boldsymbol{\theta}_o < 0] | \mathbf{x}_i\} = P(u_i < 0 | \mathbf{x}_i) = \tau$.

- Recall that $E[\mathbf{s}(\mathbf{w}_i, \theta_o)] = \mathbf{0}$ is a key condition to show asymptotic normality of M-estimators in the standard cases (twice continuously differentiable objective function with nondegenerate Hessian).
- A condition used in the asymptotic normality proof is that $\hat{\theta}$ satisfies

$$N^{-1/2} \sum_{i=1}^N \mathbf{s}_i(\hat{\theta}) = o_p(1).$$

- The remaining parts of the proof are more difficult (Newey and McFadden, *Handbook of Econometrics*).

- One difficulty is that the Hessian of the objective function, by any sensible definition, is a matrix of zeros (since the score is piecewise constant).
- Instead,
one finds the Jacobian of $E[\mathbf{s}(\mathbf{w}_i, \boldsymbol{\theta})|\mathbf{x}_i]$, which often can be assumed smooth.

- Recall that

$$\mathbf{s}_i(\boldsymbol{\theta}) = -\mathbf{x}_i'(\tau - 1[y_i - \mathbf{x}_i\boldsymbol{\theta} < 0]) = -\mathbf{x}_i'(\tau - 1[-\mathbf{x}_i(\boldsymbol{\theta} - \boldsymbol{\theta}_o) + u_i < 0])$$

$$E[\mathbf{s}_i(\boldsymbol{\theta})|\mathbf{x}_i] = -\mathbf{x}_i'[\tau - F_u(\mathbf{x}_i(\boldsymbol{\theta} - \boldsymbol{\theta}_o)|\mathbf{x}_i)],$$

- The Jacobian of $E[\mathbf{s}_i(\boldsymbol{\theta})|\mathbf{x}_i]$ is

$$\nabla_{\boldsymbol{\theta}} E[\mathbf{s}_i(\boldsymbol{\theta})|\mathbf{x}_i] = f_u(\mathbf{x}_i(\boldsymbol{\theta} - \boldsymbol{\theta}_o)|\mathbf{x}_i)\mathbf{x}_i'\mathbf{x}_i,$$

where $f_u(\cdot|\mathbf{x}_i)$ is the conditional PDF of u_i given $\mathbf{x}_i$.

- Technically, it suffices that $F_u(\cdot|\mathbf{x}_i)$ is continuously differentiable in a neighborhood of zero.
- Then $\nabla_{\boldsymbol{\theta}} E[\mathbf{s}_i(\boldsymbol{\theta}_o)|\mathbf{x}_i] = f_u(0|\mathbf{x}_i)\mathbf{x}_i'\mathbf{x}_i$.

- The matrix that appears in the asymptotic variance formula is its expectation over the distribution of $\mathbf{x}_i$:

$$\mathbf{A}_o = E[f_u(0|\mathbf{x}_i)\mathbf{x}_i'\mathbf{x}_i]. \quad ((15))$$

- We have already assumed $\mathbf{A}_o$ is nonsingular.
- Under additional regularity conditions,

$$\sqrt{N}(\hat{\theta} - \theta_o) \overset{a}{\sim} Normal(0, \mathbf{A}_o^{-1}\mathbf{B}_o\mathbf{A}_o^{-1})$$

where

$$\mathbf{B}_o \equiv E[\mathbf{s}_i(\theta_o)\mathbf{s}_i(\theta_o)'] = \tau(1 - \tau)E(\mathbf{x}_i'\mathbf{x}_i). \quad ((16))$$

- $f_u(0|\mathbf{x}) > \rho > 0$ for all $\mathbf{x}$ is often used.
- We need to (essentially) rule out $f_u(0|\mathbf{x}) = 0$, as is clear from the formula $\mathbf{A}_o = E[f_u(0|\mathbf{x}_i)\mathbf{x}_i'\mathbf{x}_i]$. (This excludes some unusual bimodal distributions.)
- For all $\mathbf{x}$, $f_u(\cdot|\mathbf{x})$ should be continuous in an interval around zero. This rules out pure discreteness in y . Often u_i is assumed to have an everywhere continuous density.

Computing Standard Errors

- Linear quantile regression:

$$Q_{\tau}(y_i|\mathbf{x}_i) = \mathbf{x}_i\boldsymbol{\theta}.$$

- Let $\hat{\boldsymbol{\theta}}$ be the QR estimator and define the quantile residuals $\hat{u}_i = y_i - \mathbf{x}_i\hat{\boldsymbol{\theta}}$, $i = 1, \dots, N$.
- A consistent estimator of $\mathbf{B}$ is simply

$$\hat{\mathbf{B}} = \tau(1 - \tau) \left(N^{-1} \sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i \right). \quad ((17))$$

- This assumes correct specification of the conditional quantile.

- Generally, a consistent estimator of $\mathbf{A}$ (Powell (1991)) is

$$\hat{\mathbf{A}} = N^{-1} \sum_{i=1}^N \frac{1[|\hat{u}_i| \leq h_N]}{2h_N} \mathbf{x}_i' \mathbf{x}_i, \quad ((18))$$

where $\{h_N > 0\}$ is a nonrandom sequence shrinking to zero as $N \rightarrow \infty$ with $\sqrt{N}h_N \rightarrow \infty$.

- For example, $h_N = aN^{-1/3}$ for any $a > 0$.
- There are rules of thumb to choose a and also data dependent methods.
- Or, we might use a smoothed density so that all residuals contribute.

- If we further assume u_i and $\mathbf{x}_i$ are independent,

$$Avar\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) = \frac{\tau(1-\tau)}{[f_u(0)]^2} [E(\mathbf{x}_i'\mathbf{x}_i)]^{-1}, \quad ((19))$$

and $Avar(\hat{\boldsymbol{\theta}})$ is estimated as

$$\widehat{Avar}(\hat{\boldsymbol{\theta}}) = \frac{\tau(1-\tau)}{[\hat{f}_u(0)]^2} \left(\sum_{i=1}^N \mathbf{x}_i'\mathbf{x}_i \right)^{-1}, \quad ((20))$$

where, say, $\hat{f}_u(0)$ is a centered histogram estimator of $f_u(0)$:

$$\hat{f}_u(0) = (2Nh_N)^{-1} \sum_{i=1}^N 1[|\hat{u}_i| \leq h_N]. \quad ((21))$$

- The estimate in (20) is commonly reported by software packages.
- If the quantile function is misspecified, the “robust” form based on (18) [with $\hat{\mathbf{B}}$ as in (17)], is not valid.
- For quantile regression, a fully robust variance estimator, which allows the quantile function to be misspecified, requires a different estimator of $\mathbf{B}$.

- Kim and White (2002) and Angrist, Chernozhukov, and Fernández-Val (2006) show

$$\hat{\mathbf{B}} = N^{-1} \sum_{i=1}^N \mathbf{s}_i(\hat{\boldsymbol{\theta}}) \mathbf{s}_i(\hat{\boldsymbol{\theta}})' = N^{-1} \sum_{i=1}^N (\tau - 1[\hat{u}_i < 0])^2 \mathbf{x}_i' \mathbf{x}_i \quad ((22))$$

is consistent for $\mathbf{B}$, and then $\widehat{Avar}(\hat{\boldsymbol{\theta}}) = \hat{\mathbf{A}}^{-1} \hat{\mathbf{B}} \hat{\mathbf{A}}^{-1} / N$, with $\hat{\mathbf{A}}$ given by (18).

- Can establish the equivalence in (22) by noting

$$\mathbf{s}_i(\hat{\boldsymbol{\theta}}) = -\mathbf{x}_i'(\tau - 1[\hat{u}_i < 0]).$$

Efficiency Calculations

- Suppose in the model

$$y = \mathbf{x}\boldsymbol{\beta} + u,$$

where $\mathbf{x}$ includes unity and u is independent of $\mathbf{x}$ with a symmetric distribution about zero. Also, $f_u(0) > 0$ and $E(u^2) < \infty$.

- Then OLS and LAD are both consistent for $\boldsymbol{\beta}$ and $\sqrt{N}$ -asymptotically normal:

$$\begin{aligned} Avar\sqrt{N}(\hat{\boldsymbol{\beta}}_{OLS} - \boldsymbol{\beta}) &= \sigma^2[E(\mathbf{x}'\mathbf{x})]^{-1} \\ Avar\sqrt{N}(\hat{\boldsymbol{\beta}}_{LAD} - \boldsymbol{\beta}) &= \frac{1}{4[f_u(0)]^2}[E(\mathbf{x}'\mathbf{x})]^{-1}. \end{aligned}$$

- $\hat{\boldsymbol{\beta}}_{LAD}$ does not need finite variance.

- We can compare asymptotic efficiency by comparing the scalars out front: σ^2 versus $1/\{4[f_u(0)]^2\}$.
- Suppose $u \sim \text{Normal}(0, \sigma^2)$.

$$\begin{aligned} f_u(0) &= 1/\sqrt{2\pi\sigma^2} \\ \frac{1}{4[f_u(0)]^2} &= \frac{2\pi\sigma^2}{4} = \left(\frac{\pi}{2}\right)\sigma^2 \approx 1.57\sigma^2 \end{aligned}$$

- This shows that LAD is asymptotically inefficient – 57% less efficient – compared with OLS when the u_i come from a normal distribution.

- Now consider the $t(\nu)$ -distribution

$$f_u(0) = \frac{\Gamma((\nu + 1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)}$$
$$\text{Var}(u) = \begin{cases} \infty, & \text{if } \nu=1,2 \\ \nu/(\nu - 2), & \text{if } \nu=3,4,\dots \end{cases}$$

- LAD is more efficient than OLS if $\nu \leq 4$.

v	OLS, σ^2	LAD, $[4f_u^2(0)]^{-1}$		notes, $[4f_u^2(0)]^{-1}$	Kurtosis
1	∞	2.465		$(\frac{1}{\sqrt{\pi}\sqrt{\pi}})^{-2}/4$	∞
2	∞	2		$(\frac{\sqrt{\pi/2}}{\sqrt{2\pi}})^{-2}/4$	∞
3	3	1.849		$(\frac{1}{\sqrt{3\pi}\sqrt{\pi/2}})^{-2}/4$	∞
4	2	1.778		$(\frac{3\sqrt{\pi/4}}{2\sqrt{\pi}})^{-2}/4$	∞
5	1.667	1.733		$(\frac{2}{\sqrt{5\pi}3\sqrt{\pi/4}})^{-2}/4$	6
6	1.5	1.707		$(\frac{15\sqrt{\pi/8}}{2\sqrt{6\pi}})^{-2}/4$	3
$\vdots$	$\vdots$	$\vdots$			$\vdots$
∞	1	1.57			0

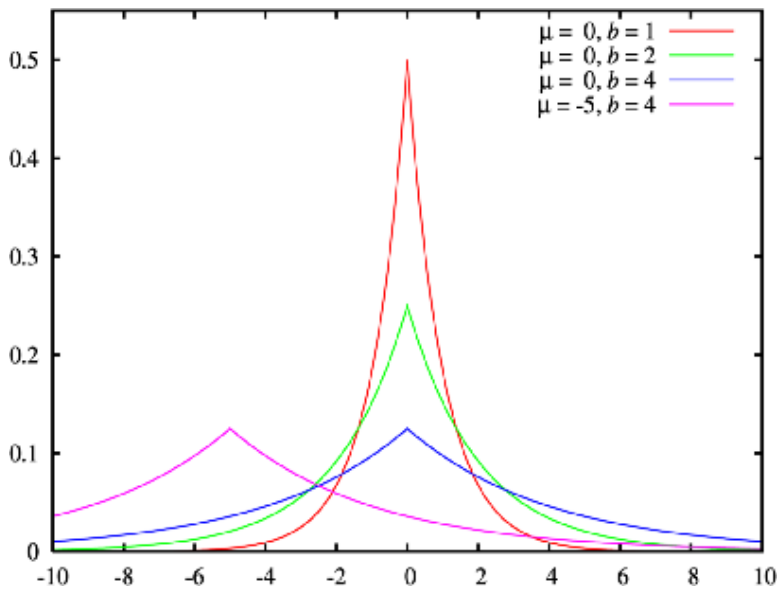
- Now suppose that u_i has a double exponential (Laplace) distribution with mean zero.
- The density is

$$f_u(x) = \frac{1}{2b} \exp(-|x|/b),$$

where $\text{Var}(u_i) = 2b^2$.

- The excess kurtosis is always 3, which implies a heavier tail than normal distribution.
- It is easily shown the MLE of β in this case is the LAD estimator. This follows because the log likelihood is

$$\mathcal{L}_N(\beta, b) = -\log(b) - b^{-1} \sum_{i=1}^N |y_i - \mathbf{x}_i \beta|.$$



- The density at zero is $f_u(0) = 1/(2b)$ and so

$$\frac{1}{4[f_u(0)]^2} = 4b^2/4 = b^2$$

whereas

$$\text{Var}(u) = \sigma^2 = 2b^2$$

- The asymptotic variance of the sample average is twice that of the sample median.

- For general asymmetric distributions it makes no sense to discuss asymptotic efficiency of the two estimators unless u is independent of $\mathbf{x}$, in which case we can compare the slopes.
- If $D(u|\mathbf{x})$ is asymmetric and depends on $\mathbf{x}$, OLS and LAD generally estimate *different* parameters.

- How about efficiencies across τ ?
- The variance takes the form

$$\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) \stackrel{a}{\sim} \text{Normal}(0, \mathbf{A}_o^{-1} \mathbf{B}_o \mathbf{A}_o^{-1})$$

where

$$\mathbf{A}_o = E[f_u(0|\mathbf{x}_i) \mathbf{x}_i' \mathbf{x}_i].$$

$$\mathbf{B}_o \equiv E[\mathbf{s}_i(\boldsymbol{\theta}_o) \mathbf{s}_i(\boldsymbol{\theta}_o)'] = \tau(1 - \tau) E(\mathbf{x}_i' \mathbf{x}_i).$$

- The efficiency depends on $\tau(1 - \tau)$:

τ	0.01	0.05	0.1	0.25	0.5
$\tau(1 - \tau)$	0.01	0.05	0.09	0.19	0.25

Fully Parametric Approaches

- If we have fully specified a model for $D(y|\mathbf{x})$ then we can learn anything about $D(y|\mathbf{x})$, including any quantile.
- If we start with

$$y = \alpha + \mathbf{x}\beta + u,$$

then specifying $D(u|\mathbf{x})$ implies a model for $D(y|\mathbf{x})$.

EXAMPLE: Assume that

$$D(u|\mathbf{x}) = \text{Normal}(0, \sigma^2 \exp(2\mathbf{x}\gamma)),$$

so that $E(y|\mathbf{x}) = \text{Med}(y|\mathbf{x}) = \alpha + \mathbf{x}\beta$.

- For any quantile $0 < \tau < 1$,

$$\text{Quant}_\tau(y|\mathbf{x}) = \alpha + \mathbf{x}\beta + \text{Quant}_\tau(u|\mathbf{x}).$$

- Write $e \equiv u / [\sigma \exp(\mathbf{x}\gamma)]$ and note that e is independent of $\mathbf{x}$ and $e \sim \text{Normal}(0, 1)$.
- But we can write

$$u = \sigma \exp(\mathbf{x}\gamma) e$$

and so

$$\begin{aligned} \text{Quant}_{\tau}(u|\mathbf{x}) &= \sigma \exp(\mathbf{x}\gamma) \text{Quant}_{\tau}(e|\mathbf{x}) = \sigma \exp(\mathbf{x}\gamma) \text{Quant}_{\tau}(e) \\ &\equiv \sigma \exp(\mathbf{x}\gamma) a_{\tau}, \end{aligned}$$

where a_{τ} is the τ^{th} quantile $N(0,1)$. (For example, if $\tau = .10$ then $a_{.10} = -1.28$.)

- It follows that

$$Quant_{\tau}(y|\mathbf{x}) = \alpha + \mathbf{x}\boldsymbol{\beta} + a_{\tau}\sigma \exp(\mathbf{x}\boldsymbol{\gamma}).$$

- If $\boldsymbol{\gamma} \neq \mathbf{0}$, this quantile function is nonlinear in $\mathbf{x}$ except when $\tau = .50$, in which case $a_{\tau} = 0$.
- This is similar to the heteroskedastic case we discussed earlier (here the scalar $\sigma \exp(\mathbf{x}\boldsymbol{\gamma})$ is always non-negative).
- When $\boldsymbol{\gamma} = \mathbf{0}$ (homoskedasticity), $Quant_{\tau}(y|\mathbf{x}) = \alpha + \mathbf{x}\boldsymbol{\beta} + a_{\tau}\sigma$ for any τ . (When $\boldsymbol{\gamma} = \mathbf{0}$, u is independent of $\mathbf{x}$, and so all quantile functions have the same slopes, $\boldsymbol{\beta}$, but with intercepts $\alpha + a_{\tau}\sigma$.)
- The parameters α , $\boldsymbol{\beta}$, σ^2 , and $\boldsymbol{\gamma}$ can be estimated by MLE using the normal density that recognizes the mean and variance both depend on $\mathbf{x}$. Two-step procedures are also available.

EXERCISE: We can drop the normality assumption in the previous example. Suppose in the equation

$$y = \alpha + \mathbf{x}\boldsymbol{\beta} + u$$

we assume

$$\begin{aligned} E(u|\mathbf{x}) &= 0 \\ \text{Var}(u|\mathbf{x}) &= h(\mathbf{x}) \end{aligned}$$

and

$$e \equiv \frac{u}{\sqrt{h(\mathbf{x})}} \text{ is independent of } \mathbf{x}.$$

- For a given τ , let η_τ be the τ^{th} quantile of e .

(i) Show that

$$Quant_{\tau}(y|\mathbf{x}) = \alpha + \mathbf{x}\beta + \eta_{\tau}\sqrt{h(\mathbf{x})}.$$

(ii) Propose consistent estimators of α and β .

(iii) If we assume $h(\mathbf{x}) = \sigma^2 \exp(2\mathbf{x}\gamma)$, how can you consistently estimate σ^2 and γ ?

(iv) Using the estimates from parts (ii) and (iii), propose a simple consistent estimator of η_{τ} for any $\tau \in (0, 1)$.